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-- Lab: Migrate Two MySQL Databases to One MSSQL Database
-- Scenario

-- You have two MySQL databases:

mysql_sales_db
mysql_hr_db

-- You want to migrate both into one Microsoft SQL Server database:

-- CompanyDW
-- Part 1: MySQL Source Database Scripts
-- 1. Create MySQL Sales Database
-- MySQL Source Database 1: Sales Database
• CREATE DATABASE mysql_sales_db;
• USE mysql_sales_db;

• CREATE TABLE customers (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_name VARCHAR(100),
    phone VARCHAR(30),
    email VARCHAR(100),
    city VARCHAR(50)
);

• CREATE TABLE products (
    product_id INT PRIMARY KEY AUTO_INCREMENT,
    product_name VARCHAR(100),
    category VARCHAR(50),
    unit_price DECIMAL(10,2)
);

• CREATE TABLE orders (
    order_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT,
    order_date DATE,
    total_amount DECIMAL(12,2),
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);
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41 • INSERT INTO customers (customer_name, phone, email, city) VALUES
42 ('Aung Aung', '0911111111', 'aung@example.com', 'Yangon'),
43 ('Su Su', '0922222222', 'su@example.com', 'Mandalay'),
44 ('Kyaw Kyaw', '0933333333', 'kyaw@example.com', 'Naypyidaw');
45
46 • INSERT INTO products (product_name, category, unit_price) VALUES
47 ('Laptop', 'Electronics', 1200000),
48 ('Mouse', 'Electronics', 25000),
49 ('Office Chair', 'Furniture', 180000);
50
51 • INSERT INTO orders (customer_id, order_date, total_amount) VALUES
52 (1, '2026-05-01', 1200000),
53 (2, '2026-05-02', 25000),
54 (3, '2026-05-03', 180000);
55 -- 2. Create MySQL HR Database
56 -- MySQL Source Database 2: HR Database
57 • CREATE DATABASE mysql_hr_db;
58 • USE mysql_hr_db;
59
60 • CREATE TABLE departments (
61     department_id INT PRIMARY KEY AUTO_INCREMENT,
62     department_name VARCHAR(100)
63 );
64
65 • CREATE TABLE employees (
66     employee_id INT PRIMARY KEY AUTO_INCREMENT,
67     employee_name VARCHAR(100),
68     email VARCHAR(100),
69     phone VARCHAR(30),
70     department_id INT,
71     salary DECIMAL(12,2),
72     hire_date DATE,
73     FOREIGN KEY (department_id) REFERENCES departments(department_id)
74 );
75

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76 • INSERT INTO departments (department_name) VALUES
77 ('IT Department'),
78 ('Finance Department'),
79 ('HR Department');
80
81 • INSERT INTO employees
82 (employee_name, email, phone, department_id, salary, hire_date) VALUES
83 ('Mg Mg', 'mgmg@example.com', '0944444444', 1, 800000, '2025-01-15'),
84 ('Hla Hla', 'hlahla@example.com', '0955555555', 2, 750000, '2025-03-10'),
85 ('Mya Mya', 'myamya@example.com', '0966666666', 3, 650000, '2025-06-20');
86

```

MySQL_DataMigration_1 customers x

Limit to 1000 rows

1 • SELECT * FROM mysql_sales_db.customers;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	customer_id	customer_name	phone	email	city
▶	1	Aung Aung	0911111111	aung@example.com	Yangon
	2	Su Su	0922222222	su@example.com	Mandalay
	3	Kyaw Kyaw	0933333333	kyaw@example.com	Naypyidaw
*	NULL	NULL	NULL	NULL	NULL

MySQL_DataMigration_1 orders x

Limit to 1000 rows

1 • SELECT * FROM mysql_sales_db.orders;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	order_id	customer_id	order_date	total_amount
▶	1	1	2026-05-01	1200000.00
	2	2	2026-05-02	250000.00
	3	3	2026-05-03	1800000.00
*	NULL	NULL	NULL	NULL


```
1  -- Part 2: MSSQL Target Database Script
2  -- 3. Create Target Database in SQL Server
3  -- MSSQL Target Database
4  CREATE DATABASE CompanyDW;
5  GO
6
7  USE CompanyDW;
8  GO
9  --4. Create Schemas for Two Source Databases
10 -- Separate schemas for migrated databases
11 CREATE SCHEMA sales;
12 GO
13
14 CREATE SCHEMA hr;
15 GO
16 --5. Create Sales Tables in MSSQL
17 CREATE TABLE sales.customers (
18     customer_id INT PRIMARY KEY,
19     customer_name NVARCHAR(100),
20     phone NVARCHAR(30),
21     email NVARCHAR(100),
22     city NVARCHAR(50)
23 );
24
25 CREATE TABLE sales.products (
26     product_id INT PRIMARY KEY,
27     product_name NVARCHAR(100),
28     category NVARCHAR(50),
29     unit_price DECIMAL(10,2)
30 );
31
32 CREATE TABLE sales.orders (
33     order_id INT PRIMARY KEY,
34     customer_id INT,
35     order_date DATE,
36     total_amount DECIMAL(12,2),
37     CONSTRAINT FK_orders_customers
38         FOREIGN KEY (customer_id)
39         REFERENCES sales.customers(customer_id)
40 );
```

```

41  --6. Create HR Tables in MSSQL
42  CREATE TABLE hr.departments (
43      department_id INT PRIMARY KEY,
44      department_name NVARCHAR(100)
45  );
46
47  CREATE TABLE hr.employees (
48      employee_id INT PRIMARY KEY,
49      employee_name NVARCHAR(100),
50      email NVARCHAR(100),
51      phone NVARCHAR(30),
52      department_id INT,
53      salary DECIMAL(12,2),
54      hire_date DATE,
55      CONSTRAINT FK_employees_departments
56          FOREIGN KEY (department_id)
57          REFERENCES hr.departments(department_id)
58  );
59

```

MySQL_DataMigration_3 x

Limit to 1000 r

```

1  -- Part 3: Migration Method
2  -- Option A: Export MySQL Data to CSV
3
4  -- In MySQL Workbench:
5
6  -- 1. Open MySQL Workbench
7  -- 2. Select mysql_sales_db
8  -- 3. Right-click table customers
9  -- 4. Select Table Data Export Wizard
10 -- 5. Export as CSV
11 -- 6. Repeat for all tables
12
13 -- Export these files:
14
15 -- customers.csv
16 -- products.csv
17 -- orders.csv
18 -- departments.csv
19 -- employees.csv
20

```

Table Data Export

Select data for export

Select source table for export: mysql_sales_db.customers

Select columns you'd like to export

Export	Column name
☒	customer_id
☒	customer_name
☒	phone
☒	email
☒	city

☒ Select / Deselect all entries

Row Offset: Count:

Advanced >>

< Back

Next >

Cancel

Table Data Export

Select output file location

Table Data Export allows you to easily export data into CSV, JSON datafiles.

File Path: C:\DE_Scholarship\customers.csv

Browse...

☒ csv ☐ json

Options:

Field Separator	,
Line Separator	CR
Enclose Strings in	"
null and NULL word as SQL keyword	YES

☒ Export to local machine

If checked, rows will be exported on the location that started Workbench.
If not checked, rows will be exported on the server.
If server and computer that started Workbench are different machines, import of that file can be done manual way only.

< Back

Next >

Cancel

customers - Excel

FileHomeInsertPage LayoutFormulasDataReview

Paste

Clipboard

A

Font

Alignment

%

Number

Conditional Form

Format as Table

Cell Styles

Styles

A1

customer_id

	A	B	C	D	E	F
1	customer_id	customer_name	phone	email	city	
2	1	Aung Aung	9.11E+08	aung@exam	Yangon	
3	2	Su Su	9.22E+08	su@exam	Mandalay	
4	3	Kyaw Kyaw	9.33E+08	kyaw@exam	Naypyidaw	

< This PC > Acer (C:) > DE_Scholarship

Search DE

Sort

View

Name	Date modified	Type	Size
customers	5/26/2026 9:08 PM	Microsoft Excel Co...	1 KB
departments	5/26/2026 9:10 PM	Microsoft Excel Co...	1 KB
employees	5/26/2026 9:11 PM	Microsoft Excel Co...	1 KB
orders	5/26/2026 9:09 PM	Microsoft Excel Co...	1 KB
products	5/26/2026 9:10 PM	Microsoft Excel Co...	1 KB


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MSSQL_DataM...Yu Ya (52)) ✎ ✕
1  |--Part 4: Import CSV into MSSQL
2  --7. Bulk Insert Sales Data
3  USE CompanyDW;
4  GO
5
6  BULK INSERT sales.customers
7  FROM 'C:\DE_Scholarship\customers.csv'
8  WITH (
9      FIRSTROW = 2,
10     FIELDTERMINATOR = ',',
11     ROWTERMINATOR = '0x0A',
12     TABLOCK
13 );
14
15 BULK INSERT sales.products
16 FROM 'C:\DE_Scholarship\products.csv'
17 WITH (
18     FIRSTROW = 2,
19     FIELDTERMINATOR = ',',
20     ROWTERMINATOR = '0x0A',
21     TABLOCK
22 );
23
24 BULK INSERT sales.orders
25 FROM 'C:\DE_Scholarship\orders.csv'
26 WITH (
27     FIRSTROW = 2,
28     FIELDTERMINATOR = ',',
29     ROWTERMINATOR = '0x0A',
30     TABLOCK
31 );
32 --8. Bulk Insert HR Data
33 BULK INSERT hr.departments
34 FROM 'C:\DE_Scholarship\departments.csv'
35 WITH (
36     FIRSTROW = 2,
37     FIELDTERMINATOR = ',',
38     ROWTERMINATOR = '0x0A',
39     TABLOCK
40 );
41
42 BULK INSERT hr.employees
43 FROM 'C:\DE_Scholarship\employees.csv'
44 WITH (
45     FIRSTROW = 2,
46     FIELDTERMINATOR = ',',
47     ROWTERMINATOR = '0x0A',
48     TABLOCK
49 );
```

```

50  --Part 5: Verify Migration
51  -- Check row counts
52  SELECT 'sales.customers' AS table_name, COUNT(*) AS total_rows FROM sales.customers
53  UNION ALL
54  SELECT 'sales.products', COUNT(*) FROM sales.products
55  UNION ALL
56  SELECT 'sales.orders', COUNT(*) FROM sales.orders
57  UNION ALL
58  SELECT 'hr.departments', COUNT(*) FROM hr.departments
59  UNION ALL
60  SELECT 'hr.employees', COUNT(*) FROM hr.employees;

```

100 %

✓ No issues found

Results Messages

	table_name	total_rows
1	sales.customers	3
2	sales.products	3
3	sales.orders	3
4	hr.departments	3
5	hr.employees	3

```

61  --Check Migrated Sales Data
62  SELECT
63      o.order_id,
64      c.customer_name,
65      c.city,
66      o.order_date,
67      o.total_amount
68  FROM sales.orders o
69  INNER JOIN sales.customers c
70      ON o.customer_id = c.customer_id;

```

100 %

✓ No issues found

Results Messages

	order_id	customer_name	city	order_date	total_amount
1	1	"Aung Aung"	Yangon	2026-05-01	1200000.00
2	2	"Su Su"	Mandalay	2026-05-02	25000.00
3	3	"Kyaw Kyaw"	Naypyidaw	2026-05-03	180000.00

```

72  --Check Migrated HR Data
73  SELECT
74      e.employee_id,
75      e.employee_name,
76      d.department_name,
77      e.salary,
78      e.hire_date
79  FROM hr.employees e
80  INNER JOIN hr.departments d
81      ON e.department_id = d.department_id;

```

100 %

✓ No issues found

Results Messages

	employee_id	employee_name	department_name	salary	hire_date
1	1	"Mg Mg"	"IT Department"	800000.00	2025-01-15
2	2	"Hla Hla"	"Finance Department"	750000.00	2025-03-10
3	3	"Mya Mya"	"HR Department"	650000.00	2025-06-20

```

82  --Part 6: Migration Validation Queries
83  -- Check orphan orders
84  SELECT o.*
85  FROM sales.orders o
86  LEFT JOIN sales.customers c
87       ON o.customer_id = c.customer_id
88  WHERE c.customer_id IS NULL;

```

100 % No issues found

Results Messages

```

90  -- Check orphan employees
91  SELECT e.*
92  FROM hr.employees e
93  LEFT JOIN hr.departments d
94       ON e.department_id = d.department_id
95  WHERE d.department_id IS NULL;

```

100 % No issues found

Results Messages

	employee_id	employee_name	email	phone	department_id	salary	hire_date
--	-------------	---------------	-------	-------	---------------	--------	-----------

```

97  -- Check NULL important fields
98  SELECT *
99  FROM sales.customers
100 WHERE customer_name IS NULL OR phone IS NULL;

```

100 % No issues found

Results		Messages			
	customer_id	customer_name	phone	email	city

```

102 SELECT *
103 FROM hr.employees
104 WHERE employee_name IS NULL OR department_id IS NULL;

```

100 % No issues found

Results Messages

employee_id	employee_name	email	phone	department_id	salary	hire_date
-------------	---------------	-------	-------	---------------	--------	-----------